

Supplementary Material for Healthcare Affordability Barriers Among US Stroke Survivors

Working supplement prepared 18 September 2026. This document accompanies the 2019–2025 NHIS manuscript. Tables contain analysis results, not illustrative values. References use the numbering in the main manuscript.

Supplementary Methods

Sources and harmonization

Seven annual Sample Adult ZIP files, seven annual income ZIP files, and the special 2020 partial Sample Adult file supplied the microdata. The source manifest also includes seven annual adult codebooks, seven survey descriptions, and seven income technical documents. It records the download URL, size, and SHA-256 for each of the 36 files. Source files were downloaded on 18 September 2026. The original planning report informed project selection; its embedded citation tokens and implementation suggestions were checked against primary sources rather than treated as governing instructions.

The questionnaire crosswalk contains the exact questions, universes, and PDF pages for core variables in each year. A separate covariate crosswalk documents education, coverage, clinical variables, smoking, and disability. Questionnaire text was compared after normalizing capitalization for the 2019 prescription-use question. A reported stroke was a prior clinician diagnosis; neither subtype nor recency was inferred.

Annual estimates retain the full 2020 adult file, including followback observations, and use WTFA_A. Pooled estimates instead retain only records linked to adultpart20 and use WTSA_P for 2020. Household identifiers are matched within year, and the partial-file HHX_2020 field is matched to the annual HHX field. All other annual weights are WTFA_A. Dividing pooled weights by seven yields an average annual target population. No population total is presented as a count of unique survivors across the seven years.

Survey designs are defined before restricting to the stroke domain. The same published PSTRAT/PPSU identifiers are used across years, following the 2025 Survey Description. Domain variance calculations retain the parent design rather than treat the selected stroke sample as a simple random sample. No finite population correction beyond the survey package defaults was supplied. A singleton PSU in the underlying design would cause the analysis to fail rather than be silently assigned an arbitrary adjustment.

Outcome classification

Outcome definitions

Measure	NHIS variables	Universe
Stroke history	STREV_A	All sample adults
Delayed medical care	MEDDL12M_A	All sample adults
Forgone medical care	MEDNG12M_A	All sample adults

Measure	NHIS variables	Universe
Forgone needed prescriptions	RXDG12M_A	All sample adults
Prescription use	RX12M_A	All sample adults
Dose skipping, dose reduction, delayed filling	RXSK12M_A; RXLS12M_A; RXDL12M_A	RX12M_A=1
Any affordability barrier	Any applicable positive among six items	Known applicability and response rules described below

Responses coded 1 and 2 indicate yes and no. Refusal, not ascertained, don't know, and missing values do not become no. Any applicable yes identifies a positive composite, even if a different component is unknown. An all-negative composite requires every applicable response to be known. For the primary composite, RX12M_A=2 is an explicit nonuse branch; the conditional questions themselves remain structurally blank. If prescription use is unknown and all universal questions are negative, the primary outcome is unknown. The prescription-underuse denominator includes only respondents with RX12M_A=1 and a classifiable underuse outcome. This distinction is essential when reproducing the analysis.

Covariate definitions

Domain	Variables	Analysis categories
Age	AGEP_A	18–85; 85 is top-coded; natural spline or 18–64/65+
Sex	SEX_A	Male; female
Race/Hispanic origin	HISPALLP_A	Seven published categories
Region	REGION	Northeast; Midwest; South; West
Education	EDUC_A; EDUCP_A	Four harmonized attainment groups
Income	RATCAT_A; IMPNUM or IMPNUM_A	<100%; 100–199%; 200–399%; ≥400% FPL
Insurance	NOTCOV_A and seven coverage indicators	Private; public without private; military only; uninsured
Clinical history	HYPEV_A; DIBEV_A; CHDEV_A	Yes/no hypertension, diabetes, coronary disease
Smoking	SMKCIGST_A	Never; former; current; indeterminate missing
Self-rated health	PHSTAT_A	Fair/poor vs good/very good/excellent
Disability	DISAB3_A	Published Washington Group disability recode

Insurance presence codes 1 and 2 both mean coverage is present; they differ in the availability of additional information. Code 3 means absence for the individual coverage indicators. The uninsured variable NOTCOV_A uses a different coding convention: 1 means not covered and 2 means covered. The hierarchy assigns private coverage first among covered respondents, then public without private coverage,

then military only when all other listed coverage indicators are explicitly absent. Ambiguous coverage is missing. Private coverage may coexist with public or military coverage, and public without private coverage may coexist with military coverage. Military-only status is therefore deliberately narrower than any military coverage.

Income uses RATCAT_A instead of rounded income-to-poverty ratios. Categories 1–3 correspond to below 100% FPL, 4–7 to 100%–199%, 8–11 to 200%–399%, and 12–14 to at least 400%. The imputation index is named IMPNUM in 2019–2020 and IMPNUM_A thereafter. All ten annual imputations were linked by year, household, and index, with duplicate-key and complete-match assertions. The point estimate in a descriptive income row is the mean of the ten completed-data weighted percentages. Fractional unweighted income counts are averages over imputations, not fractional respondents.

Estimands and models

The primary descriptive estimand is the weighted prevalence of any affordability barrier among noninstitutionalized adults reporting prior stroke and a classifiable outcome. Adjusted PRs condition on the stated covariate set. They are not causal effects, incidence ratios, or recurrence estimates. Model 1 adjusts for categorical year, age spline, sex, race/Hispanic origin, and region. Model 2 adds education, income, and insurance. Model 3 adds hypertension, diabetes, coronary heart disease, smoking, self-rated health, and disability. All three models use the same 6,777 records. The age-group sequence substitutes an age-group indicator for the age spline.

For each model, the coefficient vector is averaged over the ten imputations. Total covariance equals average within-imputation survey covariance plus 1.1 times the between-imputation coefficient covariance. Confidence intervals use the finite-sample degrees of freedom returned by `mitools::MIcombine`, supplied with the minimum complete-data residual design degrees of freedom across the fits. Poisson coefficients are exponentiated to obtain PRs. This procedure propagates income-imputation uncertainty but does not impute missing outcomes or other covariates.

For annual standardized prevalence, the fitted logistic model assigns each person to each survey year in turn while keeping their other covariates fixed. The predicted probabilities are averaged using the pooled complete-case weights. The linearized contribution includes the regression-coefficient influence multiplied by the prediction gradient and the weighted deviation of each predicted probability from its standardized mean. Their joint covariance therefore includes uncertainty in the reference population distribution and its covariance with model estimation. Before standardization, the exported regression influences were checked against the model covariance. Imputation-specific margins and their covariance were pooled on the logit scale. The 2025-minus-2019 absolute contrast was pooled directly on the probability scale.

The bounded logistic sensitivity applies the same standardization to income, insurance, disability, and the separate age-group model. Each comparison assigns the full reference sample to the comparison and reference categories and forms their prevalence ratio. These marginal estimands differ from conditional modified Poisson PRs and can involve extrapolation to covariate combinations with limited support; agreement is a model-sensitivity observation, not proof of correct specification or positivity.

The categorical-year and year-by-age-group tests use a pooled Wald F statistic based on the selected coefficients and their Rubin total covariance. The denominator degrees of freedom are the minimum scalar pooled degrees of freedom for those coefficients. This is an approximate multivariate test, not a

specialized D1 or D2 multiple-imputation test. With the observed small income-related missing-information fractions for year terms, it supplies a practical exploratory summary; the tables and intervals remain the primary description of annual variation.

Confidence intervals and precision screen

Annual observed confidence intervals are design-based logit intervals. For pooled component estimates, an effective sample size is calculated from prevalence and its Taylor variance, capped by the observed sample size and adjusted by the squared ratio of the 0.975 normal quantile to the corresponding design-t quantile. Beta quantiles based on that effective sample size provide bounded intervals. This is a documented design-adjusted beta implementation rather than a claim of exact certification against every NCHS production rule. Subgroup intervals use logit pooling across imputations; nonincome subgroup point estimates are unchanged across those imputations.

The screening rule flags a proportion when its unweighted or variance-based effective sample size is below 30, its confidence-interval width exceeds 0.30, or its width exceeds 0.05 and is more than 130% of the smaller of the estimate and its complement. Design degrees of freedom below eight prompt review. These checks are informed by NCHS guidance.[12] The unknown-insurance subgroup is flagged and its unstable prevalence is omitted from the formatted subgroup table. Main annual, component, and reported pooled subgroup estimates pass the implemented screen. This screen does not establish reliability for every regression coefficient or small race/insurance subgroup contrast.

Sensitivity analyses and implementation checks

The component models use the primary-model complete-covariate population restricted to observed values of the component, preserving a comparable adjustment sample. Their denominators therefore differ from the broader component prevalence denominators. The no-disability model omits the disability term. The working-age model retains a continuous-age spline within ages 18–64. The no-2020 model removes 2020 records and its year level; keeping the common factor of one-seventh in weights does not affect model coefficients or their linearized variance. The missing-category diagnostic retains the 7,104 observed primary outcomes and includes explicit unknown levels for incomplete nonincome factors.

All 160 modified Poisson fits converged. The full model had 14–16 fitted values above one per imputation, with maximum values approximately 1.9. The bounded logistic sensitivity used 20 additional fits, all converged. These diagnostic findings are disclosed rather than treating log-link fitted values as valid individual probabilities. The main standardized-year fits also converged.

Independent Python verification reconstructed 12 annual or pooled primary estimates and Taylor standard errors directly from the downloaded ZIPs. A separate NumPy Newton algorithm reproduced the first-imputation full modified Poisson model using the exported design matrix; maximum absolute coefficient discrepancy was below 3×10^{-13} and maximum covariance discrepancy below 3×10^{-9} . Rubin coefficient means and total standard errors were independently checked for all 16 model specifications. This verifies numerical implementation within the stated scope; it is not an independent recreation of all clinical recodes or a validation of the study's causal interpretation.

Analysis development and reporting boundaries

The project was initiated from a planning report and initial descriptive results were reviewed before completing the full covariate specification. The analysis is consequently not described as preregistered or

outcome-blinded. The report's five-imputation suggestion was corrected to ten using the actual CDC files. Employment was omitted because it overlaps strongly with age/disability and required additional harmonization choices outside the final adjustment set. A year-by-insurance interaction was not fit because the military-only group had 109 observed records and 17 events across all years. No data-driven significance rule selected the final model. Alternative handling of the ten unknown-age records and detailed disability-domain models were not undertaken and are not necessary to reproduce the presented estimates.

Supplementary tables

Table S1 Annual and pooled cohort counts

Year	Annual stroke	Annual outcome observed	Pooled stroke	Pooled outcome observed	Pooled events
2019	1201	1191	1201	1191	231
2020	1059	1051	685	679	105
2021	1010	1001	1010	1001	179
2022	1008	999	1008	999	147
2023	1131	1116	1131	1116	169
2024	1244	1229	1244	1229	215
2025	902	889	902	889	138

Annual stroke records total 7,555; pooled stroke records total 7,181. The difference comes from excluding 2020 followback records in pooled analyses. These are unweighted counts.

Table S2 Missing nonincome variables in the pooled stroke sample

Variable	Missing n	Missing %
any_barrier	77	1.07
age	0	0.00
sex	0	0.00
race	0	0.00
region	0	0.00
education	35	0.49
insurance	34	0.47
hypertension	9	0.13
diabetes	10	0.14
coronary	68	0.95
fairpoor	11	0.15

Variable	Missing n	Missing %
smoking	253	3.52
disability	0	0.00

Denominator is 7,181 stroke records. Missingness overlaps across variables; the counts must not be summed to derive complete cases. Among records with observed primary outcomes, 327 had at least one missing adjustment variable.

Table S3 Full characteristics of the pooled outcome observed sample

Characteristic	Overall N=7,104	No barrier N=5,920	Any barrier N=1,184
Age 65 years or older	4,763 (59.0)	4,237 (64.0)	526 (35.9)
Age 18–64 years	2,341 (41.0)	1,683 (36.0)	658 (64.1)
Male	3,269 (48.5)	2,777 (49.7)	492 (43.3)
Female	3,835 (51.5)	3,143 (50.3)	692 (56.7)
Hispanic, any race	599 (11.5)	478 (11.2)	121 (12.8)
Non-Hispanic White only	5,052 (66.6)	4,279 (67.6)	773 (61.7)
Non-Hispanic Black only	1,068 (15.8)	852 (14.9)	216 (19.7)
Non-Hispanic Asian only	171 (3.3)	146 (3.5)	25 (2.4)
Non-Hispanic AIAN only	77 (0.9)	66 (0.9)	11 (0.8)
Non-Hispanic AIAN with another race	91 (1.3)	63 (1.1)	28 (1.8)
Other single or multiple races	46 (0.7)	36 (0.6)	10 (0.8)
Northeast	1,032 (15.6)	901 (16.4)	131 (11.8)
Midwest	1,681 (22.0)	1,405 (22.3)	276 (20.6)
South	2,942 (43.1)	2,362 (41.0)	580 (52.7)
West	1,449 (19.4)	1,252 (20.3)	197 (15.0)
Bachelor's degree or higher	1,670 (20.1)	1,437 (20.6)	233 (17.4)
Some college or associate	2,149 (28.7)	1,743 (28.2)	406 (30.9)
High school or GED	2,202 (32.3)	1,848 (32.6)	354 (30.8)
Less than high school	1,050 (18.3)	866 (18.0)	184 (19.7)
Education unknown	33 (0.6)	26 (0.5)	7 (1.2)
Income $\geq$ 400% FPL	1,681.2 (24.2)	1,559.9 (27.0)	121.3 (11.0)
Income 200–399% FPL	2,133.3 (31.0)	1,786.9 (31.0)	346.4 (31.2)
Income 100–199% FPL	1,986.9 (27.5)	1,549.6 (25.6)	437.3 (36.0)
Income <100% FPL	1,302.6 (17.4)	1,023.6 (16.4)	279.0 (21.8)
Private, with or without public coverage	2,888 (40.3)	2,465 (40.9)	423 (37.2)

Characteristic	Overall N=7,104	No barrier N=5,920	Any barrier N=1,184
Public without private	3,871 (53.6)	3,263 (54.9)	608 (47.4)
Military only	109 (1.6)	92 (1.6)	17 (1.7)
Uninsured	207 (4.0)	80 (2.0)	127 (13.2)
Insurance unknown	29 (0.5)	20 (0.5)	9 (0.6)
No hypertension	1,747 (25.7)	1,455 (25.3)	292 (27.8)
Hypertension	5,349 (74.2)	4,458 (74.6)	891 (72.1)
Hypertension unknown	8 (0.1)	7 (0.1)	1 (0.1)
No diabetes	5,136 (71.3)	4,330 (72.3)	806 (66.9)
Diabetes	1,960 (28.6)	1,583 (27.6)	377 (33.0)
Diabetes unknown	8 (0.1)	7 (0.1)	1 (0.0)
No coronary heart disease	5,212 (74.0)	4,369 (74.8)	843 (70.8)
Coronary heart disease	1,828 (25.2)	1,500 (24.5)	328 (28.4)
Coronary heart disease unknown	64 (0.7)	51 (0.7)	13 (0.8)
Never smoked	3,196 (46.0)	2,710 (46.6)	486 (43.5)
Former smoker	2,592 (34.0)	2,240 (35.5)	352 (27.2)
Current smoker	1,132 (16.8)	821 (14.7)	311 (26.3)
Smoking unknown	184 (3.2)	149 (3.2)	35 (3.0)
Good or better self-rated health	3,607 (48.0)	3,201 (51.6)	406 (31.7)
Fair or poor self-rated health	3,486 (51.9)	2,712 (48.3)	774 (68.1)
Self-rated health unknown	11 (0.1)	7 (0.1)	4 (0.2)
Without disability	4,384 (61.2)	3,742 (62.3)	642 (56.3)
With disability	2,720 (38.8)	2,178 (37.7)	542 (43.7)

Cells are unweighted n (survey-weighted column percentage). Income counts are averages across ten imputations and may be fractional. Unknown values remain in column denominators. Categories within each variable sum to approximately 100% because of rounding. FPL, federal poverty level. AIAN, American Indian or Alaska Native. The race/Hispanic-origin categories follow the published public-use recode.

Table S4 Pooled component prevalence

Outcome	Observed n	Events	Weighted % (95% CI)
Any affordability barrier	7104	1184	17.9 (16.8–19.1)
Three universal questions	7106	1012	15.5 (14.3–16.6)
Delayed medical care	7116	554	8.5 (7.7–9.4)

Outcome	Observed n	Events	Weighted % (95% CI)
Forgone medical care	7115	541	8.4 (7.6–9.3)
Forgone needed prescriptions	7103	647	10.0 (9.1–11.0)
Medication underuse among users	6754	635	10.1 (9.2–11.1)

Components overlap. Medication-underuse estimates use prescription users only. Each prevalence denominator includes all stroke records with a classifiable corresponding outcome; it is not restricted to complete model covariates. All displayed estimates pass the implemented precision screen.

Table S5 Full Model 3 associations

Contrast	aPR (95% CI)	P value
Year 2020 vs 2019	0.89 (0.69–1.14)	0.351
Year 2021 vs 2019	0.96 (0.78–1.19)	0.702
Year 2022 vs 2019	0.84 (0.67–1.04)	0.112
Year 2023 vs 2019	0.85 (0.69–1.05)	0.140
Year 2024 vs 2019	1.06 (0.87–1.29)	0.569
Year 2025 vs 2019	0.89 (0.71–1.10)	0.274
Female vs male	1.27 (1.12–1.43)	<0.001
Non-Hispanic White only vs Hispanic	1.20 (0.95–1.51)	0.124
Non-Hispanic Black only vs Hispanic	1.14 (0.88–1.48)	0.323
Non-Hispanic Asian only vs Hispanic	0.78 (0.45–1.35)	0.367
Non-Hispanic AIAN only vs Hispanic	0.82 (0.42–1.61)	0.570
Non-Hispanic AIAN with another race vs Hispanic	1.06 (0.69–1.63)	0.800
Other single or multiple races vs Hispanic	1.26 (0.71–2.25)	0.431
Midwest vs Northeast	1.21 (0.97–1.50)	0.098
South vs Northeast	1.42 (1.15–1.75)	<0.001
West vs Northeast	1.24 (0.96–1.61)	0.095
Some college/associate vs bachelor's or higher	0.89 (0.75–1.06)	0.202
High school/GED vs bachelor's or higher	0.75 (0.62–0.89)	0.001
Less than high school vs bachelor's or higher	0.80 (0.64–0.99)	0.038
Income 200–399% vs ≥400% FPL	2.07 (1.65–2.61)	<0.001
Income 100–199% vs ≥400% FPL	2.50 (1.94–3.21)	<0.001
Income <100% vs ≥400% FPL	2.00 (1.52–2.63)	<0.001

Contrast	aPR (95% CI)	P value
Public without private vs private	0.72 (0.63–0.83)	<0.001
Military only vs private	0.81 (0.52–1.28)	0.372
Uninsured vs private	1.79 (1.49–2.16)	<0.001
Hypertension vs none	0.99 (0.86–1.14)	0.863
Diabetes vs none	1.12 (0.97–1.30)	0.126
Coronary heart disease vs none	1.37 (1.18–1.58)	<0.001
Former vs never smoking	0.98 (0.85–1.14)	0.820
Current vs never smoking	1.21 (1.04–1.42)	0.015
Fair/poor vs good or better health	1.58 (1.37–1.82)	<0.001
Disability vs none	1.19 (1.05–1.35)	0.006

The model includes an age spline with 3 degrees of freedom. Spline basis coefficients and the intercept are retained in model_coefficients.csv; they do not individually represent clinical group contrasts. No multiplicity correction was applied. Sparse race and military-coverage contrasts require caution.

Table S6 Selected associations in sensitivity analyses

Analysis	n	Income 100–199% vs ≥400% FPL	Uninsured vs private	Disability vs none
Exclude 2020	6,124	2.59 (2.00–3.36)	1.82 (1.49–2.21)	1.18 (1.04–1.34)
Age 18–64 only	2,231	2.33 (1.67–3.25)	1.85 (1.51–2.27)	1.08 (0.92–1.27)
No disability	4,209	2.50 (1.79–3.47)	1.80 (1.42–2.28)	Not applicable
Universal composite	6,776	2.45 (1.87–3.22)	2.09 (1.70–2.57)	1.26 (1.10–1.45)
Delayed medical care	6,774	2.58 (1.76–3.79)	2.69 (1.98–3.66)	1.35 (1.10–1.65)
Forgone medical care	6,776	2.46 (1.66–3.65)	2.99 (2.23–4.00)	1.40 (1.15–1.71)
Forgone prescriptions	6,775	2.52 (1.75–3.65)	2.09 (1.58–2.77)	1.24 (1.04–1.49)
Medication underuse	6,449	2.85 (2.01–4.03)	1.74 (1.32–2.31)	1.23 (1.03–1.48)
Unknown-category diagnostic	7,104	2.45 (1.91–3.14)	1.85 (1.55–2.21)	1.16 (1.03–1.31)

Cells are aPRs (95% CIs) from the corresponding fully adjusted model. No-disability models omit the disability term. The complete coefficient sets, including all income categories, are supplied as CSV. Working-age disability estimates are compatible with no association. Outcome-specific model denominators differ from Table S4 because they require complete adjustment variables.

Table S7 Bounded logistic standardized sensitivity

Contrast	Marginal PR (95% CI)	Reference %	Comparison %
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Contrast	Marginal PR (95% CI)	Reference %	Comparison %
Income 200–399% FPL vs Income $\geq$ 400% FPL	1.99 (1.59–2.49)	9.6	19.0
Income 100–199% FPL vs Income $\geq$ 400% FPL	2.41 (1.88–3.09)	9.6	23.0
Income <100% FPL vs Income $\geq$ 400% FPL	1.91 (1.45–2.51)	9.6	18.3
Public without private vs Private, with or without public coverage	0.72 (0.63–0.83)	20.1	14.5
Military only vs Private, with or without public coverage	0.83 (0.52–1.32)	20.1	16.7
Uninsured vs Private, with or without public coverage	1.99 (1.63–2.43)	20.1	40.0
With disability vs Without disability	1.18 (1.04–1.33)	16.7	19.6
Age 18–64 years vs Age 65 years or older	2.13 (1.86–2.44)	12.0	25.5

Logistic predictions remain between zero and one. Marginal PRs are pooled on the log scale; displayed prevalences are arithmetic means of imputation-specific margins, so their displayed ratio may differ slightly from the pooled PR. The age contrast uses the age-group model. These standardized estimates have a different estimand from conditional modified Poisson PRs.

Table S8 Pooled prevalence in principal subgroups

Subgroup	Observed n	Weighted % (95% CI)
Age 65 years or older	4,763	10.9 (9.8–12.0)
Age 18–64 years	2,341	28.0 (25.9–30.2)
Income $\geq$ 400% FPL	1681.2	8.2 (6.7–10.0)
Income 200–399% FPL	2133.3	18.0 (16.0–20.3)
Income 100–199% FPL	1986.9	23.5 (21.1–26.1)
Income <100% FPL	1302.6	22.5 (19.6–25.6)
Private, with or without public coverage	2,888	16.6 (14.9–18.4)
Public without private	3,871	15.8 (14.5–17.3)
Military only	109	18.2 (11.1–28.4)
Uninsured	207	58.7 (50.7–66.3)
Without disability	4,384	16.5 (15.1–17.9)
With disability	2,720	20.2 (18.4–22.1)

Income counts are averages across ten imputations. The unknown-insurance subgroup (29 records) failed the precision screen, and its prevalence is not displayed. These are unadjusted subgroup estimates.

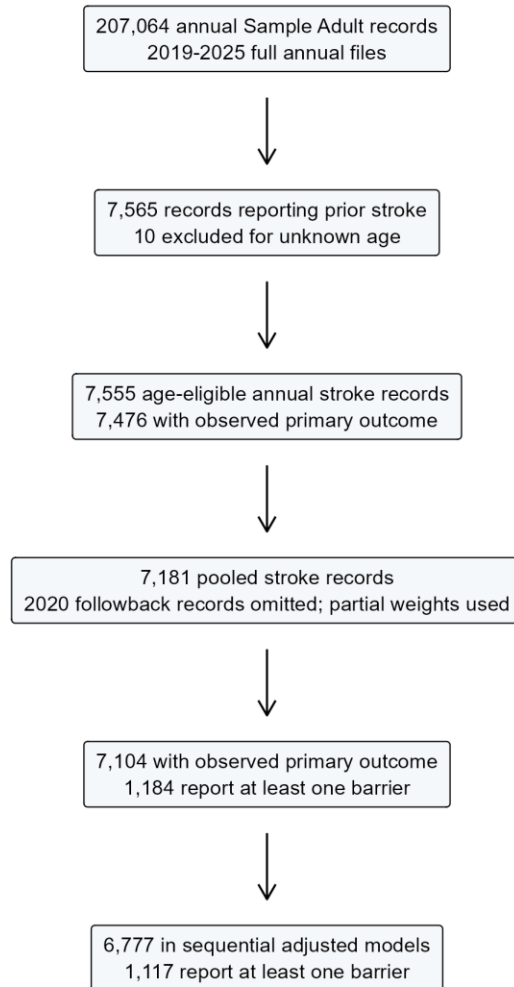
Table S9 Modified Poisson model diagnostics

Model	n	Events	Fitted >1 per imputation	Largest fitted value	Convergence
M1	6777	1117	0–0	0.517	All converged
M2	6777	1117	3–4	1.190	All converged
M3	6777	1117	14–16	1.943	All converged
Age_M1	6777	1117	0–0	0.466	All converged
Age_M2	6777	1117	3–4	1.106	All converged
Age_M3	6777	1117	15–17	1.912	All converged
AgeInteraction	6777	1117	17–19	1.888	All converged
No2020	6124	1016	12–16	1.943	All converged
WorkingAge	2231	620	14–15	2.127	All converged
NoDisability	4209	607	5–6	1.802	All converged
Universal	6776	955	13–17	1.972	All converged
Delayed	6774	523	3–5	1.518	All converged
ForgoneCare	6776	509	4–8	1.651	All converged
ForgoneRx	6775	610	4–6	1.488	All converged
RxUnderuse	6449	598	5–6	1.709	All converged
MissingCategory	7104	1184	16–19	1.891	All converged

Ten fits per specification. Fitted log-link values above one are not interpreted as individual probabilities. The bounded logistic sensitivity supports the major associations; it does not establish perfect specification. Model labels correspond to the accompanying script and coefficient CSV.

Supplementary Figure 1 Analytic sample selection

Selection of annual and pooled analytic samples



Annual and pooled samples use different 2020 inclusion and weighting rules. The 7,555 annual stroke records include the full 2020 sample; the 7,181 pooled stroke records exclude 2020 followback records. The model sample additionally requires an observed outcome and complete nonincome adjustment variables. Annual counts are record counts, not unique persons across years.

Draft STROBE reporting checklist

STROBE cross sectional checklist

Item	Location	Coverage
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Item	Location	Coverage
1	Title and abstract	Design, population, period and quantitative findings supplied.
2–3	Introduction	Background, prior studies and objectives stated.
4–5	Methods / Study design	Repeated cross-sectional design, public NHIS setting and 2019–2025 period.
6	Methods / Population; Figure S1	Eligibility, stroke definition, age exclusion, annual vs pooled selection.
7–8	Methods / Outcomes and covariates	Definitions, universes and cross-year source verification; crosswalk files included.
9	Discussion / Limitations	Self-report, nonresponse, institutional exclusion, confounding and missingness addressed.
10	Methods / Population	All eligible records in the selected years; no formal power-based sample selection.
11	Methods / Covariates	Age spline and categorical income/education definitions given.
12	Methods / Analysis; Supplementary Methods	Survey weighting, MI, missingness, interaction and sensitivity procedures.
13	Results / Sample; Table S1; Figure S1	Counts and reasons for exclusions; different annual and pooled paths.
14	Table 1; Tables S2–S3	Characteristics and missingness; income counts averaged across imputations.
15	Table 2; Tables S4 and S8	Outcome counts and annual, component and subgroup estimates.
16	Table 3; Table S5; Results	Adjusted PRs, covariate sets, precision and absolute annual difference.
17	Sensitivity Results; Tables S6–S9	Sensitivity estimates and model diagnostics.
18–21	Discussion and Conclusions	Key results, limitations, cautious interpretation and generalizability.
22	Declarations	Author contributions and competing-interests statement supplied; no funding statement is included at the direction of the authors.

This is a draft reporting map, not certification of submission readiness. Final page numbers and author declarations can be added after journal adaptation.

Additional data files

Machine-readable aggregate files include all model coefficients and intervals, per-imputation coefficients and covariance matrices, Table 1 counts and weighted percentages, annual and pooled prevalences, standardized contrasts, model diagnostics, missingness, and the complete source manifest. The working package also includes the exact codebook crosswalks and source excerpts. Public microdata are retained separately in the project workspace and can be reacquired with the supplied downloader. Author-specific declarations remain clearly identified in the main manuscript and author information sheet.